

STEERING ANGLE DETECTION

Description:

This system uses two Adafruit Feather nRF52840 Express boards, each paired with an ICM-42688 IMU. There is one unit on the wheel measuring angular velocity ($d\theta/dt$) and wirelessly sends via 2.4 GHz RF link to an identical unit on the dash, which holds a fixed reference IMU. Comparing the moving and stationary IMUs **removes gyro bias** that builds up over time, isolating the steering wheel's true rotation... **$\theta(t)$** ... its angle over time.

Each Feather is powered by a Li-Po battery with onboard charging, voltage monitoring on A6, and a low-battery LED on D5 (see pinout).

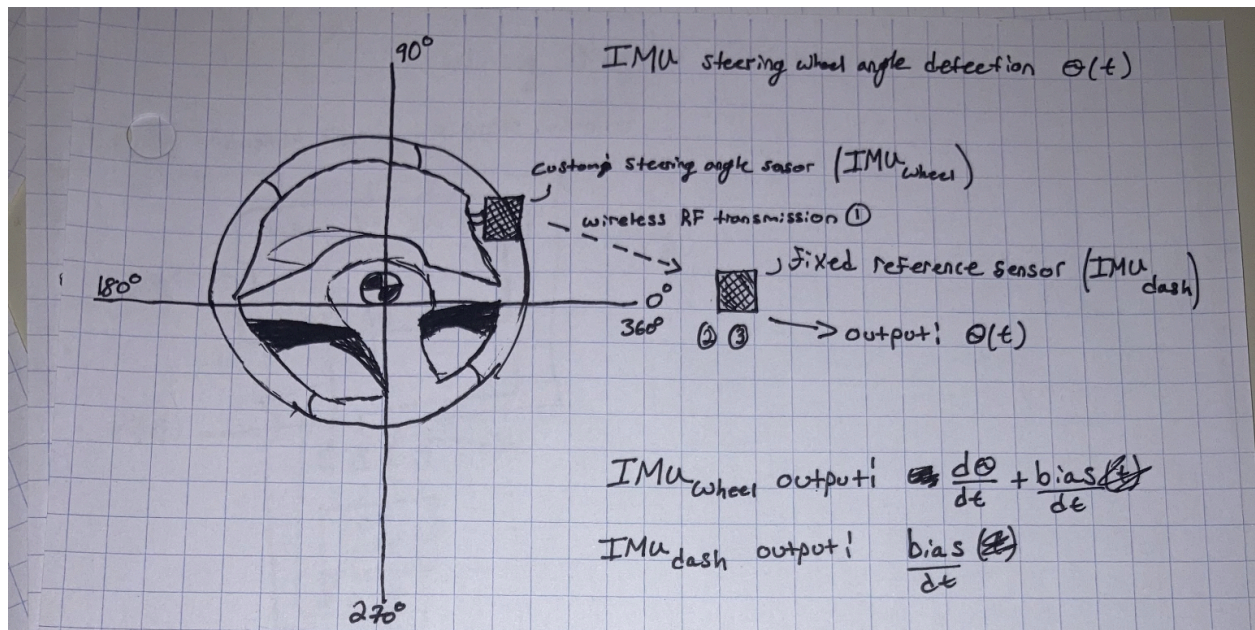
Order of Operation:

1. Wheel Unit output = $d\theta / dt + \text{bias}$
2. Wheel Unit transmits its output to the Dash Unit
3. Dash Unit output = (fixed θ does not change with time) + bias
4. Computation: $(d\theta / dt + \text{bias}) - \text{bias} = d\theta / dt$
5. Integration: $d\theta / dt = \theta(t)$
6. Dash Unit sends **$\theta(t)$** to PC or Goggles via bluetooth, 2.4 GHz RF, or USB-C (all are available on the Adafruit Feather).

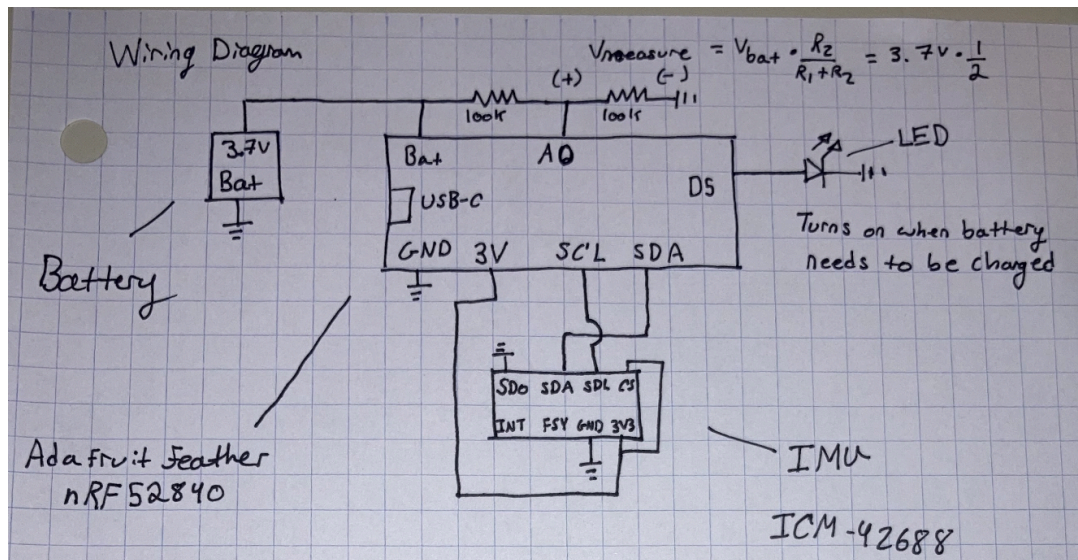
HARDWARE (two of everything):

1. Adafruit Feather nRF52840 x 2
2. ICM-42688 x 2
3. 3.7 LiPo Battery x 2
4. LED / resistors / perfboard (in stock)
5. Custom 3d printed enclosure
6. Velcro strap to attach to the steering wheel / dash...

DIAGRAM:



PINOUT of DASH / Wheel Unit (exact same hardware)



- **A0 (analog input 0)** measures the battery voltage. We use a voltage divider because the max voltage of analog pins are 3.2V... Since we are monitoring the battery voltage, we can turn on a LED tied to **D5 (digital output 5)** when the battery is low... indicating we need to charge the system with usb-c.
- **SDA (Serial Data):** The line that carries data between the microcontroller and the IMU in the I²C protocol. It's used for both sending and receiving bits of information.
- **SCL (Serial Clock):** The line that provides timing pulses so both devices know when to read or write each bit on SDA.
- **CS (Chip Select):** A control line used in SPI communication; when pulled high (3.3 V) it disables SPI and tells the ICM-42688 to use I²C instead.
- **SDO (Serial Data Out):** In SPI mode it outputs data from the IMU, but in I²C mode it instead sets the sensor's address (tied to GND = 0x68 or to 3 V = 0x69).